

Why do we need tensors?

Scalars:

describe a quantity at a point

e.g. Temperature

Vectors:

describe quantity and a direction

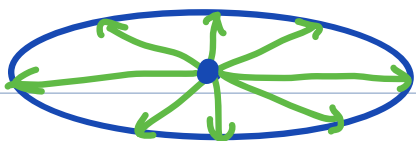
e.g. velocity (speed + direction)



Tensors:

describes how a quantity changes with direction

Think of an ellipsoid



Examples: anisotropic properties
stress, strain

moment of inertia

Second-order Tensors

Linear operators : $\underline{v} = \underline{A} \underline{u}$

maps vector $\underline{u} \in \mathcal{V}$ into vector $\underline{v} \in \mathcal{V}$

Two tensors $\underline{A}$ and $\underline{B}$ are equal if

$$\underline{A} \underline{v} = \underline{B} \underline{v} \quad \text{for all } \underline{v} \in \mathcal{V}$$

Zero tensor: $\underline{0} \underline{v} = \underline{0}$ for all $\underline{v} \in \mathcal{V}$

Identity tensor: $\underline{I} \underline{v} = \underline{v}$ for all $\underline{v} \in \mathcal{V}$

Basic algebra

α = scalar, $\underline{v}$ = vector, $\underline{A}$ & $\underline{B}$ 2nd-ord. tensors

1) $(\alpha \underline{A}) \underline{v} = \underline{A} (\alpha \underline{v})$ scalar multiplication

2) $(\underline{A} + \underline{B}) \underline{v} = \underline{A} \underline{v} + \underline{B} \underline{v}$ tensor sum

3) $(\underline{A} \underline{B}) \underline{v} = \underline{A} (\underline{B} \underline{v})$ tensor product

4) (tensor scalar product $\rightarrow$ later)

1 + 2 $\Rightarrow$ imply linearity

1, 2, 3 produce other tensors

set $\mathcal{V}^2$ of second order tensors $\Rightarrow$ vector space

Q: What is a basis for $\mathcal{V}^2$?

Representation of a tensor

In a frame $\{\underline{e}_i\}$ a second order tensor

$\underline{\underline{S}}$ is represented by nine numbers

$$S_{ij} = \underline{e}_i \cdot \underline{\underline{S}} \underline{e}_j$$

Matrix representation of tensor in $\{\underline{e}_i\}$

$$[\underline{\underline{S}}] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix} \in \mathbb{R}^3 \times \mathbb{R}^3$$

Note that $[\underline{\underline{S}}]_{ij} = S_{ij}$

Consider $\underline{v} = \underline{\underline{S}} \underline{u}$ where $\underline{v} = v_k \underline{e}_k$, $\underline{u} = u_j \underline{e}_j$

$$v_k \underline{e}_k = \underline{\underline{S}} (u_j \underline{e}_j) = \underline{\underline{S}} \underline{e}_j u_j$$

multiply by $\underline{e}_i$ from left

$$v_k \underline{e}_i \cdot \underline{e}_k = \underline{e}_i \cdot \underline{\underline{S}} \underline{e}_j u_j$$

$$v_k \delta_{ik} = \underline{e}_i \cdot \underline{\underline{S}} \underline{e}_j u_j$$

$$v_i = (\underline{e}_i \cdot \underline{\underline{S}} \underline{e}_j) u_j$$

$$v_i = S_{ij} u_j$$

Dyadic Product

The dyadic product of two vectors $\underline{a}$ and $\underline{b}$ is the 2nd-order tensor $\underline{a} \otimes \underline{b}$ defined by

$$(\underline{a} \otimes \underline{b}) \underline{v} = (\underline{b} \cdot \underline{v}) \underline{a} \quad \text{for all } \underline{v} \in \mathcal{V}$$

This has the form: $\underline{A} \underline{v} = \alpha \underline{a}$

in components: $A_{ij} v_j = \alpha a_i$

$$\alpha = \underline{b} \cdot \underline{v} = b_j v_j$$

$$A_{ij} = [\underline{a} \otimes \underline{b}]_{ij}$$

$$\Rightarrow [\underline{a} \otimes \underline{b}]_{ij} v_j = b_j v_j a_i$$

$$[\underline{a} \otimes \underline{b}]_{ij} v_j = (a_i b_j) v_j$$

$$\Rightarrow [\underline{a} \otimes \underline{b}]_{ij} = a_i b_j$$

So that

$$[\underline{a} \otimes \underline{b}] = \begin{bmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{bmatrix} = \underline{a} \underline{b}^T$$

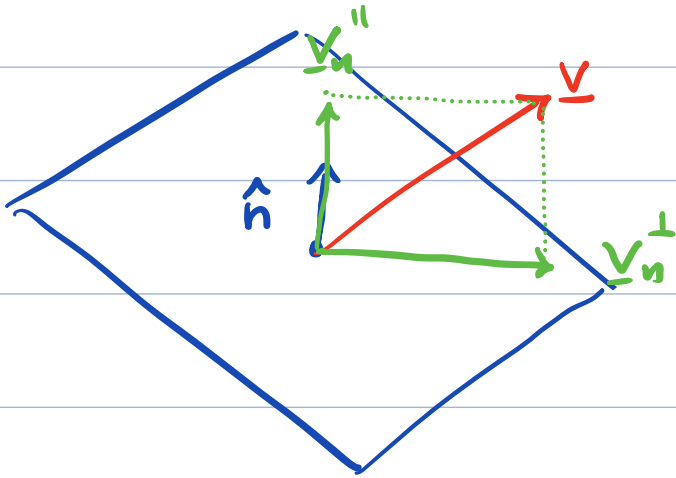
Linearity of dyadic product:

for scalars $\alpha, \beta \in \mathbb{R}$ and vectors $\underline{a}, \underline{b}, \underline{v}, \underline{w} \in V$

$$(\underline{a} \otimes \underline{b})(\alpha \underline{v} + \beta \underline{w}) = \alpha (\underline{a} \otimes \underline{b}) \underline{v} + \beta (\underline{a} \otimes \underline{b}) \underline{w}$$

Projection tensors

commonly used to partition forces on a surface.



$$\underline{v} = \underline{v}_n^{\parallel} + \underline{v}_n^{\perp}$$

Use dot product:

$$\underline{v}_n^{\parallel} = (\underline{v} \cdot \hat{n}) \hat{n}$$

$$\underline{v}_n^{\perp} = \underline{v} - \underline{v}_n^{\parallel}$$

Tensors ? $\underline{v}_n^{\parallel} = \underline{P}_n^{\parallel} \underline{v}$
 $\underline{v}_n^{\perp} = \underline{P}_n^{\perp} \underline{v}$

use dyadic property !

$$\underline{v}_n^{\parallel} = (\underline{v} \cdot \hat{n}) \hat{n} = (\hat{n} \otimes \hat{n}) \underline{v} = \underline{P}_n^{\parallel} \underline{v}$$

$$\underline{v}_n^{\perp} = \underline{v} - (\hat{n} \otimes \hat{n}) \underline{v} = (\underline{I} - \hat{n} \otimes \hat{n}) \underline{v} = \underline{P}_n^{\perp} \underline{v}$$

Projection tensors:

$$\underline{P}_n^{\parallel} = \hat{n} \otimes \hat{n}$$

$$\underline{P}_n^{\perp} = \underline{I} - \hat{n} \otimes \hat{n}$$

Basis for V^2

Given any frame $\{\underline{e}_i\}$ the nine dyadic products $\{\underline{e}_i \otimes \underline{e}_j\}$ form a basis for V^2 .

Any second-order tensor $\underline{\underline{S}}$ can be written as linear combination

$$\underline{\underline{S}} = S_{ij} \underline{e}_i \otimes \underline{e}_j$$

where $S_{ij} = \underline{e}_i \cdot \underline{\underline{S}} \underline{e}_j$

Consider $\underline{v} = \underline{\underline{S}} \underline{u}$ with $\underline{v} = v_i \underline{e}_i$, $\underline{u} = u_k \underline{e}_k$

$$v_i \underline{e}_i = S_{ij} (\underline{e}_i \otimes \underline{e}_j) (u_k \underline{e}_k)$$

$$= S_{ij} u_k (\underline{e}_i \otimes \underline{e}_j) \cdot \underline{e}_k \quad \text{apply def. of dyadic}$$

$$= S_{ij} u_k (\underline{e}_j \cdot \underline{e}_k) \underline{e}_i = S_{ij} u_k \delta_{kj} \underline{e}_i$$

$$v_i \underline{e}_i = S_{ij} u_j \underline{e}_i$$

Index notation for tensor-vector multiplication

$$v_i = S_{ij} u_j \quad \text{used often}$$

Note: Transfer property of Kronecker delta

$$v_i = \delta_{ij} u_j = u_i$$

also applies to indices of tensors

for example above

$$S_{ij} u_k \delta_{kj} e_i =$$

$$u_k \underbrace{S_{ij} \delta_{kj}}_{S_{ik}} e_i = S_{ik} u_k e_i$$

The product of two dyadic products

$$(\underline{a} \otimes \underline{b})(\underline{c} \otimes \underline{d}) = (\underline{b} \cdot \underline{c}) \underline{a} \otimes \underline{d} \Rightarrow \text{HW2}$$

needed for tensor product.

Tensor algebra in components

Addition: $\underline{\underline{H}} = \underline{\underline{S}} + \underline{\underline{T}}$

$$H_{ij} \underline{e}_i \otimes \underline{e}_j = S_{ij} \underline{e}_i \otimes \underline{e}_j + T_{ij} \underline{e}_i \otimes \underline{e}_j \\ = (S_{ij} + T_{ij}) \underline{e}_i \otimes \underline{e}_j$$

$$\boxed{H_{ij} = S_{ij} + T_{ij}}$$

Scalar multiplication: $\underline{\underline{H}} = \alpha \underline{\underline{S}} \Rightarrow \boxed{H_{ij} = \alpha S_{ij}}$

Product: $\underline{\underline{H}} = \underline{\underline{S}} \underline{\underline{T}}$

$$\underline{\underline{H}} = S_{ij} (\underline{e}_i \otimes \underline{e}_j) T_{kl} (\underline{e}_k \otimes \underline{e}_l) \\ = S_{ij} T_{kl} \underbrace{(\underline{e}_i \otimes \underline{e}_j)(\underline{e}_k \otimes \underline{e}_l)}_{\text{product of two dyads}}$$

$$= S_{ij} T_{kl} (\underline{e}_j \otimes \underline{e}_k) \underline{e}_i \otimes \underline{e}_l \\ \delta_{jk}$$

$$= S_{ij} T_{jl} \underline{e}_i \otimes \underline{e}_l$$

$$H_{il} \underline{e}_i \otimes \underline{e}_l = S_{ij} T_{jl} \underline{e}_i \otimes \underline{e}_l$$

$\Rightarrow$

$$\boxed{H_{il} = S_{\underset{j}{j}} T_{\underset{j}{j}l}}$$

note the dummy j !

Determinant and Inverse

The determinant of $\underline{\underline{A}} \in \mathbb{R}^2$ is the scalar

$$\det(\underline{\underline{A}}) = \det \begin{vmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{vmatrix} = \epsilon_{ijk} [\underline{\underline{A}}]_{i1} [\underline{\underline{A}}]_{j2} [\underline{\underline{A}}]_{k3}$$

where $[\underline{\underline{A}}]_{i1}$, $[\underline{\underline{A}}]_{j2}$, $[\underline{\underline{A}}]_{k3}$ are the columns of $[\underline{\underline{A}}]$

Triple scalar product: $(\underline{a} \times \underline{b}) \cdot \underline{c} = \det[\underline{a}, \underline{b}, \underline{c}] = \epsilon_{ijk} a_i b_j c_k$
determinants $\Rightarrow$ volumes

Properties: $\det(\underline{\underline{A}} \underline{\underline{B}}) = \det(\underline{\underline{A}}) \det(\underline{\underline{B}})$

$$\det(\underline{\underline{A}}^T) = \det(\underline{\underline{A}})$$

$$\det(\alpha \underline{\underline{A}}) = \alpha^n \det(\underline{\underline{A}}) \quad (\underline{\underline{A}} \text{ is } n \times n)$$

$\underline{\underline{A}}$ is singular if $\det \underline{\underline{A}} = 0$.

If $\det \underline{\underline{A}} \neq 0$ then the inverse $\underline{\underline{A}}^{-1}$ exists

$$\underline{\underline{A}}^{-1} \underline{\underline{A}} = \underline{\underline{A}} \underline{\underline{A}}^{-1} = \underline{\underline{I}}$$

Transpose of a tensor

To any $\underline{\underline{S}} \in \mathcal{V}^2$ we associate a transpose $\underline{\underline{S}}^T \in \mathcal{V}^2$ the unique tensor such that

$$\underline{\underline{S}} \underline{u} \cdot \underline{v} = \underline{u} \cdot \underline{\underline{S}}^T \underline{v} \quad \text{for all } \underline{u}, \underline{v} \in \mathcal{V}$$

This implies that $S_{ij}^T = S_{ji}$ as follows

$$(S_{ij} u_j e_i) \cdot (v_l e_l) = (u_k e_k) \cdot (S_{ij}^T v_j e_i)$$

$$S_{ij} u_j v_l (e_i \cdot e_l) = S_{ij}^T v_j u_k (e_k \cdot e_i)$$

$$S_{ij} u_j v_l \delta_{il} = S_{ij}^T v_j u_k \delta_{ki}$$

$$S_{ij} u_j v_i = S_{ij}^T v_j u_i$$

$$S_{ij} u_j v_i = S_{ji}^T u_j v_i$$

$$\Rightarrow S_{ij} = S_{ji}^T \quad \checkmark$$

rename indices
 $i \leftrightarrow j$ on rhs

Properties of transpose:

$$(\underline{\underline{A}}^T)^T = \underline{\underline{A}}$$

$$(\underline{\underline{A}} \underline{\underline{B}})^T = \underline{\underline{B}}^T \underline{\underline{A}}^T$$

$$(\underline{u} \otimes \underline{v})^T = \underline{v} \otimes \underline{u}$$

$\underline{\underline{S}}$ is symmetric if $\underline{\underline{S}} = \underline{\underline{S}}^T$ $S_{ij} = S_{ji}$

$\underline{\underline{S}}$ is skew-symmetric if $\underline{\underline{S}} = -\underline{\underline{S}}^T$ $S_{ij} = -S_{ji}$

Symmetric - Skew decomposition:

Any tensor $\underline{\underline{S}} \in \mathcal{V}^2$ can be written as

$$\begin{aligned}\underline{\underline{S}} &= \underline{\underline{E}} + \underline{\underline{W}} \\ \underline{\underline{E}} &= \frac{1}{2}(\underline{\underline{S}} + \underline{\underline{S}}^T) & \underline{\underline{E}} &= \underline{\underline{E}}^T \\ \underline{\underline{W}} &= \frac{1}{2}(\underline{\underline{S}} - \underline{\underline{S}}^T) & \underline{\underline{W}} &= -\underline{\underline{W}}^T\end{aligned}$$

Note: Skew tensors often related to rotation

Trace of a tensor

We define the trace of a dyad as

$$\text{tr}(\underline{a} \otimes \underline{b}) = \underline{a} \cdot \underline{b} = a_i b_i$$

this implies that

$$\text{tr}(\underline{\underline{A}}) = A_{ii} = A_{11} + A_{22} + A_{33}$$

as follows $\text{tr}(\underline{A}_{ij} \underline{e}_i \otimes \underline{e}_j) = A_{ij} \text{tr}(\underline{e}_i \otimes \underline{e}_j)$
 $= A_{ij} \delta_{ij} = A_{ii}$

Properties: $\text{tr}(\underline{A}^T) = \text{tr}(\underline{A})$

$$\text{tr}(\underline{A}\underline{B}) = \text{tr}(\underline{B}\underline{A})$$

$$\text{tr}(\underline{A} + \underline{B}) = \text{tr}(\underline{A}) + \text{tr}(\underline{B})$$

$$\text{tr}(\alpha \underline{A}) = \alpha \text{tr}(\underline{A})$$

Decomposition: $\underline{A} = \alpha \underline{I} + \text{dev } \underline{A}$

Spherical tensor: $\alpha \underline{I}$ where $\alpha = \frac{1}{3} \text{tr}(\underline{A})$

Deviatoric tensor: $\text{dev } \underline{A} = \underline{A} - \alpha \underline{I}$

$$\text{tr}(\text{dev } \underline{A}) = 0$$